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(2) According to Newton's third law, there are two reaction forces exerted by the book.

(i) The book exerts an equal and opposite force (mg) on the Earth which acts upwards.

(ii) The book exerts a force which is equal and opposite to normal force on the surface of the table (N) acting downwards.

Note

It is to be emphasized that while applying Newton's third law it is wrong to conclude that the book on the table is at rest due to the downward gravitational force exerted by the Earth and the equal and opposite reacting normal force exerted by the table on the book. Action and reaction forces never act on the same body.

(3) The free body diagram of the book is shown in the figure.

EXAMPLE 3.2

If two objects of masses 2.5 kg and 100 kg experience the same force 5 N, what is the acceleration experienced by each of them?

Solution

From Newton's second law (in magnitude form), F = ma

For the object of mass 2.5 kg, the acceleration is $a = \frac{F}{m} = \frac{5}{2.5} = 2\ m\ s^{-2}$

For the object of mass 100 kg, the acceleration is $a = \frac{F}{m} = \frac{5}{100} = 0.05\ m\ s^{-2}$

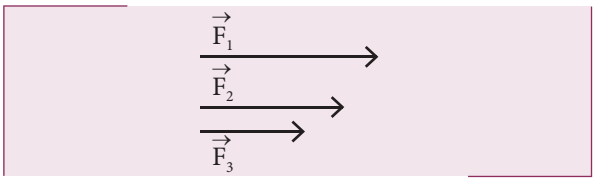
Note

Even though the force applied on both the objects is the same, acceleration experienced by each object differs. The acceleration is inversely proportional to mass. For the same force, the heavier mass experiences lesser acceleration and the lighter mass experiences greater acceleration.

When an apple falls, it experiences Earth's gravitational force. According to Newton's third law, the apple exerts equal and opposite force on the Earth. Even though both the apple and Earth experience the same force, their acceleration is different. The mass of Earth is enormous compared to that of an apple. So an apple experiences larger acceleration and the Earth experiences almost negligible acceleration. Due to the negligible acceleration, Earth appears to be stationary when an apple falls.

EXAMPLE 3.3

Which is the greatest force among the three force $\vec{F}_1, \vec{F}_2, \vec{F}_3$ shown below



Solution

Force is a vector and magnitude of the vector is represented by the length of the vector. Here $\vec{F}_1$ has greater length compared to other two. So $\vec{F}_1$ is largest of the three.